

INTRODUCTION TO P-ADIC NUMBERS

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1. INTRODUCTION

Say we want to investigate if some equation has integer solutions or not, reducing mod n for some $n \in \mathbb{N}$ is a good idea because we might find that there are no solutions mod n and thus no integer solutions.

Example 1. There are no integer solutions of $3x^3 + y^2 = 2$, because reducing mod 3 we get $y^2 = 2$ and calculating squares mod 3 we see that 2 is not a square.

On the other hand, if we found all solutions mod n , then we know the possible residues mod n an integer solution will have and so we made progress towards finding an integer solution.

Suggestion: Take a prime p and start by finding a solution mod p . Afterwards we can try to "lift" it to a solution mod p^2 and so on, thus we construct a sequence of "approximations" to an integer solution.

Example 2. Let's look for a solution of $x^2 = 2$ using $p = 7$ and the strategy stated above. looking mod 7 we compute squares and by exhaustion find that $3^2 \equiv 9 \equiv 2 \pmod{7}$, so pick 3 as a solution mod 7. Now we want to look for a solution mod $7^2 = 49$ and we know there is a solution of the form $3 + 7k$ for $k \in \{0, \dots, 6\}$ (these are the classes of $\mathbb{Z}/49\mathbb{Z}$ that mod 7 reduce to 3), so plug it in and solve for k :

$$\begin{aligned}(3 + 7k)^2 &\equiv 2 \pmod{49} \\ \Leftrightarrow 9 + 42k + 49k^2 &\equiv 2 \pmod{49} \\ \Leftrightarrow 7 + 42k &\equiv 0 \pmod{49} \\ \Leftrightarrow 1 + 6k &\equiv 0 \pmod{7} \\ \Leftrightarrow k &= 1\end{aligned}$$

Note that it is not a coincidence that everything was divisible by 7, as we started with a solution mod 7, this will happen every time. Furthermore, we will always get to a linear equation in k because the quadratic term has coefficient divisible by the modulus. Thus, we can continue and construct solutions mod 7^n for every $n \in \mathbb{N}$ that are "lifts" of the previous solutions (i.e when reducing mod 7^{n-1} we get the previous solution).

Well, that's a problem, $x^2 = 2$ obviously doesn't have an integer solution. We have constructed something completely different than an integer solution.

1.1. p-adic integers.

Definition 3. Given a prime $p \in \mathbb{N}$, a p -adic integer is a sequence $(a_n)_{n=1}^{\infty}$ such that $a_n \in \mathbb{Z}/p^n\mathbb{Z}$ and $a_n \equiv a_m \pmod{p^m}$ for every $m \leq n$. We denote the set of p -adic integers by $\mathbb{Z}_p$ and endow it the structure of a ring with coordinate-wise operations.

Remark. Note that $\mathbb{Z}$ embeds into $\mathbb{Z}_p$ via the diagonal map $a \mapsto (a, a, a, \dots)$

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We formalize the notion of approximation we intuitively used in the previous example by constructing the p -adic valuation and absolute value:

Definition 4. The p -adic valuation $v_p: \mathbb{Z} \rightarrow \mathbb{Z} \cup \{\infty\}$ is defined by $v_p(a) := \max\{n \in \mathbb{N} \mid p^n \mid a\}$ and then extended to $\mathbb{Q}$ by $v_p(a/b) := v_p(a) - v_p(b)$ (check this is well defined!)

Claim 5. *The p -adic valuation has the following properties:*

- (1) $v_p(x) = \infty \Leftrightarrow x = 0$.
- (2) $v_p(xy) = v_p(x) + v_p(y)$.
- (3) $v_p(x + y) \geq \min\{v_p(x), v_p(y)\}$.
- (4) If $v_p(x) \neq v_p(y)$ then $v_p(x + y) = \min\{v_p(x), v_p(y)\}$.

Definition 6. We define the p -adic absolute value on $\mathbb{Q}$ by $|x|_p = p^{-v_p(x)}$.

From the previous claim we get a corresponding claim about the p -adic absolute value

Claim 7. *The p -adic absolute value has the following properties:*

- (1) $|x|_p = 0 \Leftrightarrow x = 0$.
- (2) $|xy|_p = |x|_p |y|_p$.
- (3) *Non-archimedean triangle inequality (stronger than the regular one):* $|x + y|_p \leq \max\{|x|_p, |y|_p\}$.
- (4) If $|x|_p > |y|_p$ then $|x + y|_p = |x|_p$

Now Similarly to the definition of $\mathbb{R}$ as the completion of $\mathbb{Q}$ with respect to the usual euclidean absolute value, we can do the same.

Definition 8. Denote by $\mathbb{Q}_p$ the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$. Note that (as always) the absolute value extends to the completion (and so does v_p).

Remark. Note that the completion of $\mathbb{Z}$ with respect to $|\cdot|_p$ is $\mathbb{Z}_p$ (this is not strictly true because the completion of $\mathbb{Z}$ contains equivalence classes of integer sequences, but they are "naturally" isomorphic), so $\mathbb{Z}_p \subseteq \mathbb{Q}_p$. As a matter of fact, we have

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\} \qquad \mathbb{Q}_p = \text{Frac}(\mathbb{Z}_p)$$

Claim 9. *Every element of $\mathbb{Q}_p$ has a unique representation as $\sum_{i=r}^{\infty} a_i p^i$ where $r \in \mathbb{Z}$ and $a_i \in \{0, \dots, p-1\}$. This is called the p -adic expansion of the element.*

Sketch of Proof. If $(x_i)_{i=1}^{\infty} = x \in \mathbb{Z}_p$, then we can look "mod p^i " by looking at x 's i -th coordinate. We construct the p -adic expansion inductively by looking mod p to get $x_1 \equiv a_0 \pmod{p}$, then looking mod p^2 we get $x_2 \equiv a_0 + a_1 \cdot p \pmod{p^2}$ so we can solve for a_1 , etc. For a general $x \in \mathbb{Q}_p$, note that for large enough N , $p^N \cdot x \in \mathbb{Z}_p$, so we can go as before and then divide by p^N , to get terms with negative indices. $\square$

So now we have a nice normed field and we can try to do geometry/topology/analysis. there is a lot to say about it so we just mention a few funny anecdotes for a normed space with a non-archimedean triangle inequality:

- (1) Every triangle in is an isosceles triangle.
- (2) Every point in a ball is "the" center of the ball.

Claim 10. *Given $\{a_n\}_{n=1}^{\infty} \subseteq \mathbb{Q}_p$, we have $\sum_{n=1}^{\infty} a_n < \infty \Leftrightarrow a_n \xrightarrow{n \rightarrow \infty} 0$*

Corollary 11. $1 + 2 + 4 + 8 + 16 + \dots = -1$ (Well... at least in $\mathbb{Z}_2$)

Proof. By the previous claim, the series $\sum_{n=0}^{\infty} 2^n$ converges in $\mathbb{Z}_2$. Denote its limit by S . Now we can do the vague manipulations you can find on the internet:

$$\begin{aligned} S &= 1 + 2 + 4 + 8 + \dots = 1 + 2(1 + 2 + 4 + 8 + \dots) = 1 + 2S \\ \Rightarrow S &= -1 \end{aligned}$$

□

Going back to our original problem, we can formalize now the idea we started with about approximating solutions:

Theorem 12 (Hensel's Lemma). *If $f \in \mathbb{Z}_p[x]$ and $a \in \mathbb{Z}_p$ satisfies*

$$f(a) \equiv 0 \pmod{p} \qquad f'(a) \not\equiv 0 \pmod{p}$$

then there is a unique $\alpha \in \mathbb{Z}_p$ such that $f(\alpha) = 0$ and $\alpha \equiv a \pmod{p}$

The condition of being a simple root is exactly what we used when we did it by hand in the beginning, every time we plugged in we got a linear equation in our parameter, which has a solution.

You can think of it as $a \in \mathbb{Z}/p^n\mathbb{Z}$ for some n by taking a representative and then mapping diagonally to $\mathbb{Z}_p$.

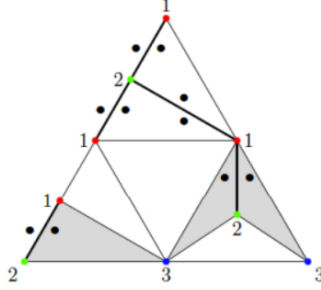
The idea of the proof is very nice and is actually just Newton's method from calculus/numerical analysis (which is surprising), so we encourage anyone interested to go check it out.

2. MONSKY'S THEOREM: AN APPLICATION

Theorem 13 (Monsky 1970). *A square cannot be dissected into an odd number of triangles with equal area.*

We will prove this theorem with the use of the 2-adic absolute value, but first we need two ingredients from combinatorics/finite geometry and Algebra

2.1. Sperner's Lemma. Let P be a polygon, each of its vertices we mark with 1, 2 or 3. Assume we are given a triangulation of P .



Definition 14. If an edge has vertices marked $i, j \in \{1, 2, 3\}$ we call that edge an ij -edge.

Definition 15 (complete triangle). We say that a triangle is complete if its vertices are marked 1, 2 and 3.

Example 16. The grey triangles in the figure above.

Claim 17 (Sperner's Lemma). *The parity of the number of complete triangles is equal to the parity of the number of 12-edges on the boundary of P . (i.e there is equality mod 2)*

Proof. Draw a dot on each side of every 12-edge (as seen in the figure above). Now we will count the number of dots in the interior of P in two ways:

- (1) Each complete triangle has exactly one dot inside it. Every other kind of triangle has either 0 or 2 dots inside of it. So the number of dots inside P has the same parity as the number of complete triangles.
- (2) A 12-edge in the interior of P contributes 2 dots and A 12-edge on the boundary of P contributes 1 dot. So the number of dots inside P has the same parity as the number of 12-edges on the boundary of P .

□

2.2. Chevalley's theorem. Without going into any details, we state the following claim which is a specific case of a theorem of Chevalley.

Claim 18. *We can extend the 2-adic absolute value on $\mathbb{Q}$ to $\mathbb{R}$.*

This is very non-constructive so after choosing such an extension, we can't say a lot about the absolute value of an irrational number. Nonetheless this is handy in the middle of computations and if in the end we get to a statement about the 2-adic absolute value of an element $\mathbb{Q}$ then we can say something.

2.3. Proof of Monsky's Theorem. We partition $\mathbb{R}^2$ as follows:

$$\begin{aligned} P_1 &:= \{(x, y) \in \mathbb{R}^2 \mid |x|_2 < 1, |y|_2 < 1\} \\ P_2 &:= \{(x, y) \in \mathbb{R}^2 \mid |x|_2 \geq 1, |x|_2 \geq |y|_2\} \\ P_3 &:= \{(x, y) \in \mathbb{R}^2 \mid |y|_2 \geq 1, |y|_2 > |x|_2\} \end{aligned}$$

Note that this is actually a partition of $\mathbb{R}^2$, as can be seen by the following sketch:

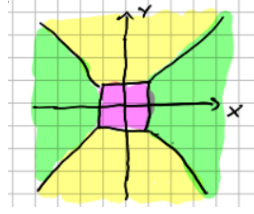


FIGURE 1. A confusing diagram, the scale is with respect to $|\cdot|_2$! Purple= P_1 , Green= P_2 , Yellow= P_3

Claim 19. *If a triangle T is complete with respect to the markings 1, 2, 3 defined by P_1, P_2, P_3 respectively, then we have*

$$|\text{Area}(T)|_2 > 1$$

Proof. Firstly, note that from the rule $|a|_2 > |b|_2 \Rightarrow |a + b|_2 = |a|_2$ we get that P_2 and P_3 are invariant under translation by elements of P_1 , so translating by P_1 preserves completeness of triangles. Secondly, as usual, the area is invariant under translation. Thus, we can assume without loss of generality that one of T 's vertices is $(0, 0)$. Denote the other vertices by $(x_2, y_2) \in P_2$ and $(x_3, y_3) \in P_3$ and we have

$$\text{Area}(T) = \frac{1}{2} \left| \det \begin{pmatrix} x_2 & x_3 \\ y_2 & y_3 \end{pmatrix} \right|_\infty = \frac{|x_2 y_3 - x_3 y_2|_\infty}{2}$$

But by construction $|x_2|_2 \geq |y_2|_2$ and $|y_3|_2 > |x_3|_2$, so multiplying we get $|x_2 y_3|_2 > |x_3 y_2|_2$ and this implies that $|x_2 y_3 - x_3 y_2|_2 = |x_2 y_3|_2$. Overall we get

$$|\text{Area}(T)|_2 = \left| \frac{1}{2} \right|_2 \cdot |x_2 y_3 - x_3 y_2|_2 = 2 |x_2 y_3|_2 = 2 \cdot |x_2|_2 \cdot |y_3|_2 \geq 2$$

□

Now we can really start the proof of Monsky's theorem

Proof. Without loss of generality, assume that the square in question is the unit square $[0, 1]^2$ and assume it is triangulated into m triangles with equal area. We mark the vertices of the triangulation by the partition P_1, P_2, P_3 we defined already. The only edge of the square that can contain a 12-edge is the bottom edge – On the other edges either $x = 0$, $x = 1$ or $y = 1$ constantly. Moreover, there are no vertices with marking 3 on the bottom edge of the square, the left corner is marked 1 and the right corner is marked 2 so necessarily there is an odd number of 12-edges on the bottom edge of the square (and those on the boundary in general).

From Sperner's lemma we get that there are an odd number of complete triangles in the triangulation, so in particular there is a complete triangle in the triangulation. Denote it's area by A , so we saw that $|A|_2 > 1$. It is a triangulation by m triangles with equal area, so we get that the overall area of the unit square (which is 1) is equal to $m \cdot A$, in particular we get

$$1 = |1|_2 = |m \cdot A|_2 = |m|_2 \cdot |A|_2 > |m|_2$$

Implying that m is even. □